

SynthLight: Portrait Relighting with Diffusion Model by Learning to Re-render Synthetic Faces

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Figure 1. SynthLight relights portraits using environment map lighting. By learning to re-render synthetic human faces, it produces realistic illumination on real portraits, such as distinct cast shadows on the neck and natural specular highlights on the skin. Despite training solely on synthetic headshots, it generalizes to half-body portraits and full-body figurines.

Abstract

We introduce *SynthLight*, a diffusion model for portrait relighting. We frame image relighting as a re-rendering problem, where pixels are transformed in response to changes in environmental lighting. Using a physically-based rendering engine, we create a dataset to simulate this lighting-conditioned transformation with 3D head assets under varying lighting. We propose training and inference strategies to bridge the gap between the synthetic and real image domains: (1) multi-task training leveraging real human portraits without lighting labels; (2) an inference time diffusion sampling scheme based on classifier-free guidance leveraging the input portrait to better preserve details. Our method generalizes to diverse real portraits and produces realistic illumination effects, including specular highlights and cast shadows, while preserving identity. Our quantitative experiments on light stage data demonstrate results comparable to state-of-the-art relighting methods. Our qualitative results on in-the-wild images showcase rich and unprecedented illumination effects.

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1. Introduction

Lighting is fundamental to portrait photography, yet manipulating it after capture remains challenging. Recent advances in generative imaging models have shown promise for controlling lighting in existing images [14, 18, 34, 58, 60], but these typically require labeled training data. For portrait relighting specifically, the most effective results have come from training on light stage data—portraits rendered with linear combinations of one-light-at-a-time (OLAT) captures. While powerful, light stage setups are constrained by physical limitations in light source density and require specialized artificial lighting equipment. In contrast, 3D workflows in VFX and gaming have long treated lighting as a straightforward endeavor through modern physically based rendering engines, where lighting control is nearly arbitrary. To relight a rendering, artists simply adjust the lighting configurations and re-render the scene.

Given a scene S and lighting L_1 , we denote the rendering as $I_1 = f_r(S, L_1)$. The inverse graphics problem aims to find S from I_1 : $S = f_{inv}(I_1, L_1)$ with known or unknown lighting. To relight rendering I_1 to I_2 under lighting L_2 , one aims to compute: $I_2 = f_r(S, L_2)$. Given only I_1 , a relight-

ing procedure seeks: $I_2 = f_r(f_{inv}(I_1), L_2) = f_{re}(I_1, L_2)$, where f_{re} is the relighting/re-rendering function. Previous approaches [21, 28, 57] have tackled this problem through inverse graphics, explicitly or implicitly, by estimating lighting-invariant intrinsic scene representations such as depth, surface normals, and albedo. This imposes limitations on subsequent rendering functions and fails to capture complex illumination effects like inter-reflections, occlusion shadows and subsurface scattering. In this paper, we propose bypassing inverse rendering by learning the relighting function using physically based 3D renderings of human heads. We render pairs of portrait images in Blender (Cycles) (I_1, L_1) and (I_2, L_2) and train a diffusion model to directly learn to “re-render” I_2 from I_1 and L_2 .

However, this approach introduces an inevitable domain gap between 3D renders and real photographs. To address this challenge, we leverage a latent diffusion model pre-trained on vast internet images for text-to-image generation. We propose to finetune the network with our face renderings and introduce simple yet effective schemes to narrow the gap between training data and in-the-wild images. During training, we propose **multi-task** training that incorporates in-the-wild images without ground truth relighting information. This allows the model to learn relighting from our synthetic dataset while maintaining knowledge of the real image domain, preventing distributional drift. We further observe that input portraits contain rich textural information. Leveraging the flexibility of diffusion model inference, we design an **inference adaptation** scheme that effectively preserves input portrait details in the relit result.

We evaluate our methods on in-the-wild portrait images, demonstrating highly detailed illumination effects that accurately capture interactions between the portrait scene and lighting. Our results produce realistic cast shadows and specular highlights on the skin. For the first time, we demonstrate an end-to-end system capable of non-trivial lighting effects including catch lights in eyes, subsurface scattering in ears, and inter-reflections with clothing. Notably, despite training only on simple headshot renderings of 3D faces without accessories, facial hair, or hats, our network generalizes to complex portrait images, including half-body shots and multi-person photographs.

We quantitatively evaluate our method on a test set of our synthetic faces dataset and on a light stage OLAT dataset. Despite using no light stage data for training, our method achieves comparable or superior results to state-of-the-art portrait relighting methods trained on OLAT data. User studies show our results are preferred across all evaluated aspects: perceptual lighting accuracy, identity preservation, and overall image quality.

We summarize our contributions as follows:

1. We propose modeling portrait relighting as a task of learning to re-render a portrait scene in 3D. Using phys-

ically based renderings of human heads under varying lighting conditions, we train a diffusion model to learn pixel transformations conditioned on lighting.

2. We introduce two techniques enabling synthetic data learning while minimizing domain gap with real images, through a training-time multi-task strategy that incorporates real images through a text-to-image task, and an inference-time approach based on classifier-free guidance that preserves portrait details in the relit result.
3. Through extensive qualitative and quantitative evaluations, we demonstrate state-of-the-art portrait relighting results, achieving high-quality lighting effects previously unattainable by existing methods.

2. Related Work

2.1. Portrait Relighting

Portrait relighting has been explored in both 2D [18, 21, 28, 29, 34, 44, 47, 53, 58, 60] and 3D [3, 6, 33, 49, 50, 52, 62], with 2D image-based methods being more relevant to our work. Due to the under-constrained nature of 2D portrait relighting, various priors have been proposed, including morphable 3D face models [4, 43], explicit inverse rendering [2, 41], and style transfer [42].

Recently, deep learning methods [27, 47] trained on light stage data [9] have advanced the state-of-the-art. Total Relighting [28] is a widely adopted physics-guided method that decomposes images into intrinsic components conditioned on HDR environment lighting [8]. However, this approach has two limitations. First, its BRDF-based reflectance model [7, 31] neglects other significant light transport modes such as subsurface scattering prevalent in certain skin types [11, 22, 26], and struggles with albedo estimation in cases involving face accessories, inter-reflections, and face paint [21, 57]. Second, light stage setups inherently limit captured lighting complexity due to restricted intensity and resolution [47, 48], affecting the modeling of complex effects such as subsurface scattering.

Motivated by these constraints, we employ diffusion models for face relighting, without predefined appearance assumptions. We use synthetic datasets rendered with physically-based methods to provide paired supervision. Our approach can synthesize advanced illumination effects in human portraits, such as hard cast shadows, subsurface scattering, and inter-reflections.

2.2. Diffusion Models for Relighting

Diffusion models [1, 10, 19, 20, 35, 37, 38, 45, 46] have become standard for text-to-image generation, image translation, and appearance editing. Their scalability on large datasets and adaptability of pretrained weights [17, 35, 59] make them particularly effective. Additionally, flexible sampling mechanisms can enhance image quality [15, 24].

Recent diffusion-based relighting methods include Di-LightNet [58], which provides fine-grained control of object lighting but relies on error-prone scene reconstruction [51], and Neural Gaffer [18], which utilizes HDR environment maps for object relighting. For portraits, Relightful Harmonization [34] and IC-Light [60] train on high-quality data (light stage captures, Objaverse renders, composited shadow materials) for background-harmonized outputs, using background as lighting condition. Diffusion Face Relighting [32] employs conditional diffusion implicit models with spherical harmonic lighting but is limited to tight facial crops. In contrast, we approach portrait relighting by learning to re-render synthetic faces. Leveraging pretrained models and our multi-task training strategy, we preserve facial priors, while classifier-free guidance [15] on the input portrait enhances textures and details in the relit results.

2.3. Domain Adaptation

Training naively on synthetic data creates a domain gap for in-the-wild portraits, harming identity preservation and photo-realism. Prior diffusion-based domain adaptation methods [12, 17, 36, 56, 59] primarily target style transfer or specific editing, not relighting.

DiffRelight [14] trains personalized diffusion models per subject, preserving identity but requires light stage capture and dedicated per-subject training. Other approaches use real data to reduce the synthetic-to-real gap: Switch-Light [21] pre-trains a masked-autoencoder [13] on real images before light stage training to learn visual features essential for relighting (e.g., structure, color, texture); Relightful Harmonization [34] bootstraps from a light stage trained model to pseudo-label in-the-wild images, then finetunes on these labels; IC-Light [60] employs extensive data augmentation; and Lumos [57] finetunes its albedo-prediction branch on real images but struggles with face paint, accessories, or strong shadows.

We propose a multi-task scheme combining text-to-image and relighting, enabling diffusion model training with real and synthetic images. Our classifier-free guidance inference preserves input portrait details. Our user study demonstrates superior visual quality, identity preservation, and lighting compared to existing methods.

3. Method

Given a portrait image I captured under unknown lighting, our goal is to synthesize a relit portrait I_R under target lighting specified by a panoramic environment map E . The relit portrait I_R should (1) preserve the subject’s identity and facial characteristics from I ; (2) reflect the lighting defined by E ; and (3) maintain photorealism. We first simulate this re-rendering in Blender to build a synthetic portrait dataset.



Figure 2. Subjects are rendered under various lighting conditions. We present two examples, each showing a subject rendered with two different environment maps. SynthLight learns to re-render synthetic faces, transforming a subject rendered with one environment map to another.

3.1. Synthetic Data for Relighting

We build a 3D human portrait generation pipeline similar to FaceSynthetics [54]. Our system begins with artist-created 3D head meshes, enhanced with detailed facial components, including eyes, teeth, gums, and hair. We augment these base models through rigging for pose variation and blend-shape deformation for diverse facial expressions. For realistic rendering, we use high-quality PBR texture maps, including albedo, normal, roughness, specular, and subsurface scattering. We combine the head with random clothing meshes to build portrait scenes using Blender and render images with Cycles.

To train our networks, we render images (samples in Fig. 2) at 512×512 resolution from 350 subjects, each with roughly 10 varied appearances, including different hairstyles, skin tones, expressions, clothes, and poses. Each sample is rendered with 10 random HDR environment maps, rotated evenly 36 times from a random initial rotation. In total, the dataset contains approximately 1.26 million images. See supplemental material for more samples.

3.2. Modeling Relighting with Diffusion Model

We build on top of Stable Diffusion [35], a text-to-image foundation model pretrained with vast internet data. As shown in Fig. 3, we incorporate the input portrait I and the target environment map E to the input of the network backbone by expanding the number of channels in the first convolutional layer of the Unet as per Palette [37], and we train the diffusion model to generate I_R , the relit portrait.

To generate training samples (I, E, T, I_R) , where T is a text prompt, we render portrait images from a subject with different HDR maps to obtain portraits n portraits $I_1 \dots I_n$. We use an off-the-shelf image captioning model [25] to caption these images. Training samples are constructed by sampling two indices $i, j \in \{1 \dots n\}$, using them to select input portrait, environment map, text prompt and target portrait as (I_i, E_j, T_j, I_j) . We use the sample to supervise our diffusion model in the following manner. First, we convert the HDR environment map into LDR by tone-mapping similar to Neural Gaffer [18]. The LDR environment map along with the input and target portraits are

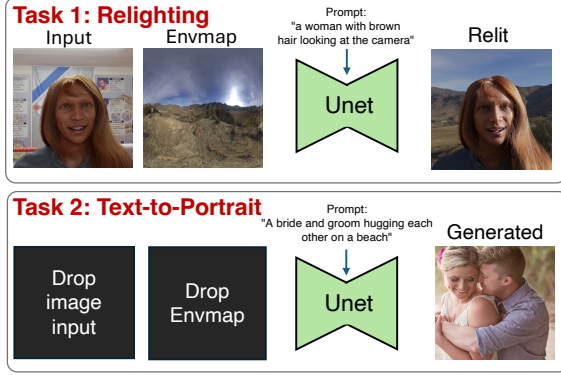


Figure 3. SynthLight training pipeline. We first train the diffusion backbone using synthetic relighting tuples (Task 1, top row; Sec.3.2). To reduce the synthetic-to-real domain gap, we jointly train on a text-to-image task (Task 2, bottom row; Sec.3.3). Our model is built on LDM [35], comprising a Unet and a VAE (omitted here for simplicity).

encoded using the encoder Enc of Stable Diffusion’s VAE, i.e., $\hat{I}_i = \text{Enc}(I_i)$, $\hat{E}_j = \text{Enc}(E_j^{\text{LDR}})$, $\hat{I}_j = \text{Enc}(I_j)$.

Following the DDPM formulation [16], we randomly sample Gaussian noise ϵ and a diffusion timestep t to add noise to the relit image latent $\hat{I}_j$ to obtain the noised latent $\hat{I}_j^t$. We concatenate $\hat{I}_i, \hat{E}_j, \hat{I}_j^t$ along the channel axis and feed it to the Unet. We train ϵ_θ with the DDPM objective:

$$\min_{\theta} \mathbb{E}_{x \in \text{Enc}(I_R), t, \epsilon \in \mathcal{N}(0, I)} \|\epsilon_\theta(x_t, I, E, T) - \epsilon\| \quad (1)$$

3.3. Multitask Training

Fine-tuning a diffusion model on synthetic data creates a domain gap when applied to in-the-wild images, degrading output quality. For instance, the model often fails to reproduce critical details, such as textures in clothing, jewelry and accessories, which are absent in the synthetic data distribution (e.g., as seen in the *Base* result in Fig. 9).

To address this, we propose multitask training to mitigate distribution drift toward synthetic renderings. Similar strategies have been applied in the context of inpainting [55] to compensate for limited training data diversity.

Specifically, we incorporate a text-to-portrait generation task that constrains the diffusion model to produce realistic portrait images from input prompts. This task is trained alongside the original relighting task. As both tasks share the same network architecture, we simply replace the image and LDR inputs with two black images, as shown in Fig. 3.

For text-to-portrait training, we curate a subset of human portraits from the LAION [40] dataset, filtered with a face detector. Details on detection and filtering are provided in the supplementary material. During training, we set the sampling ratio of synthetic to real data to 0.7 and 0.3, respectively. Incorporating real images significantly improves

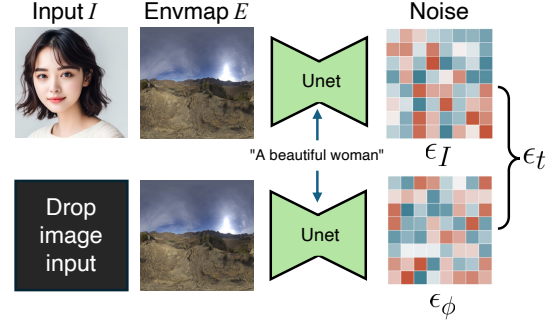


Figure 4. We apply image-conditioned classifier-free guidance during inference to balance between identity preservation and relighting effects, computing the score estimate as per Eq. (2).

identity preservation and photorealism. This aligns with findings from Relightful Harmonization [34], where a bootstrapped dataset enhances generalization in image harmonization, highlighting the importance of data diversity.

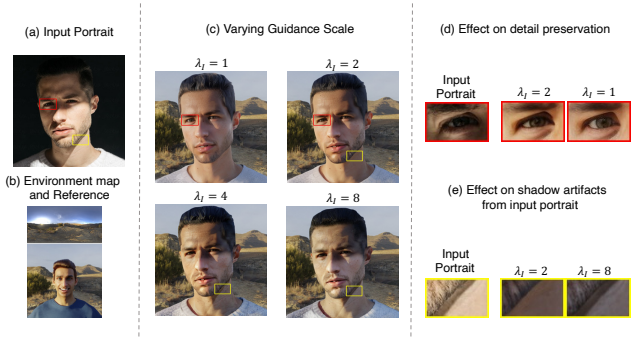


Figure 5. Effect of portrait guidance λ_I : (a) Input portrait. (b) Lighting and Blender-rendered reference. (c) Outputs with varying λ_I . (d) $\lambda_I = 1$ (no Inference Adaptation) distorts eye shape (red boxes). (e) Higher λ_I cause lighting from the input (such as shadows) to leak into the output (yellow boxes).

3.4. Inference Adaptation

We employ a simple, effective inference scheme that balances identity preservation and relighting strength. Inspired by dual-conditioning classifier-free guidance [15] from InstructPix2Pix [5], we define a similar concept in our inference. As shown in Fig. 4, at each diffusion step, the diffusion score combines score estimates from both image-conditioned and unconditioned model evaluations, where for unconditional evaluation, we drop the input image while keeping the LDR and text prompt unchanged.

Formally, the score at timestep t is:

$$\begin{aligned} \epsilon_t = & \epsilon_\theta(x_{t+1}, \phi, E, \phi) \\ & + \lambda_T(\epsilon_\theta(x_{t+1}, I, E, T) - \epsilon_\theta(x_{t+1}, I, E, \phi)) \\ & + \lambda_I(\epsilon_\theta(x_{t+1}, I, E, \phi) - \epsilon_\theta(x_{t+1}, \phi, E, \phi)). \end{aligned} \quad (2)$$

	Ours	SwitchLight [21]	IC-Light [60]	DiLightNet [58]	Neural Gaffer [18]
Test Synthetic					
LPIPS $\downarrow$	0.063	0.088	0.108	0.128	0.102
SSIM $\uparrow$	0.945	<u>0.911</u>	0.874	0.860	0.900
PSNR $\uparrow$	29.57	21.43	20.28	22.99	<u>25.33</u>
FaceNet $\downarrow$	0.165	<u>0.198</u>	0.284	0.333	0.357
Test Light Stage					
LPIPS $\downarrow$	<u>0.165</u>	0.141	0.172	0.245	0.196
SSIM $\uparrow$	<u>0.813</u>	0.853	0.789	0.703	0.788
PSNR $\uparrow$	19.70	20.30	17.44	16.62	19.31
FaceNet $\downarrow$	<u>0.173</u>	0.152	0.195	0.576	0.247

Table 1. We compare against baselines on a held-out set of our synthetic dataset (Test Synthetic) and data rendered through a light stage (Test Light Stage). While trained only on synthetic data, our model performs comparably to SwitchLight [21], a commercial relighting method trained with light stage data.

	IC-Light[60]	SwitchLight[21]	Neural Gaffer[18]
Lighting	92%	56%	65%
Quality	57%	64%	73%
Identity	52%	70%	65%

Table 2. User preference rates indicate how often SynthLight was preferred over baselines; e.g., our method was preferred 92% of times in terms of *lighting* over IC-Light [60]. SynthLight outperforms baselines in *lighting*, *image quality*, and *subject identity* as all preference rates exceed 50%.

λ_T and λ_I are guidance parameters: λ_T controls text adherence, while λ_I controls portrait guidance strength. Using these scores, we compute latents via DDIM [46]: $x_t = \text{DDIM}(x_{t+1}, \epsilon_t)$. Empirically we choose $\lambda_I \in [2, 3]$ balances detail, identity preservation, and relighting quality.

Fig. 5 shows that smaller λ_I improves relighting but reduce visual quality and facial details, while larger λ_I better preserve identity but cause lighting information, such as input shadows, to leak into outputs.

4. Experiments

4.1. Setup and Metrics

We create three test sets for evaluating our method: (a) 300 light stage rendered relighting pairs, (b) a held out subset of our synthetic faces dataset consisting of 500 images, (c) in-the-wild portraits for qualitative evaluation of visual quality. We use quantitative metrics such as SSIM, PSNR, LPIPS [61] to evaluate image fidelity and face embedding distance from FaceNet [39] to evaluate identity preservation. We train on the entire synthetic dataset but withhold 20% of the environment maps to create the light stage test set. We also hold out 10% of the subjects and 10% of the environment maps for the synthetic test set, ensuring they remain unseen during training.

4.2. Implementation details

We implement our model in PyTorch [30] and train on $32 \times 40\text{GB}$ A100 GPUs. We use a batch size of 192, a learning rate of 10^{-5} , and the Adam optimizer [23]. Training (including ablations) runs for 40K steps, taking about 1 day.

Model Initialization We initialize our model using either the IC-Light checkpoint [60] or the Stable Diffusion 1.5 checkpoint [35]. Although both initializations yield visually similar results on in-the-wild data and our light stage test set (Fig. 10), the model initialized from the Stable Diffusion 1.5 checkpoint achieves slightly lower scores on both the synthetic and light stage test sets (Tab. 3). Consequently, results in this paper are from our model initialized with the IC-Light checkpoint, unless otherwise specified.

4.3. Evaluation Results

We compare our method to state-of-the-art approaches in portrait harmonization [60], portrait relighting [21], and object relighting [18, 58], evaluating both quantitatively (see Tab. 1) on the synthetic and light stage test sets, and qualitatively (see Fig. 7 and Fig. 8). Quantitative evaluation shows that our method outperforms baselines on the synthetic test set and performs comparably to state-of-the-art portrait relighting methods like SwitchLight on the light stage test set. While our results do not always achieve the highest PSNR, they exhibit better visual relighting quality than baselines. In Fig. 7, we show the input image (left), the environment map (top), and the corresponding 3D model rendered in Blender (bottom). In Fig. 8, the left side shows the environment map (top) and input portrait (bottom).

We conduct a user study (see Tab. 2) to evaluate perceptual preference in relighting. We select 40 held-out stock portraits, each paired with a randomly sampled environment map from a test set of 200. Each portrait is relit under 12 evenly spaced rotations of the environment map, generating 480 relit images per method. For the study, we randomly sample 30 image pairs per participant, each comparing our method to a baseline. Participants answer three 2-alternative forced-choice questions assessing (1) **lighting**, (2) **image quality**, and (3) **identity preservation**. We collect 482 responses from 20 participants. Results show our method is consistently preferred across all criteria. See the supplementary for screenshots illustrating our user study.

4.4. Ablations

We conduct an ablation study to evaluate the contribution of our two key domain adaptation methods: Multi-task training (see Sec. 3.3) and inference adaptation (see Sec. 3.4).

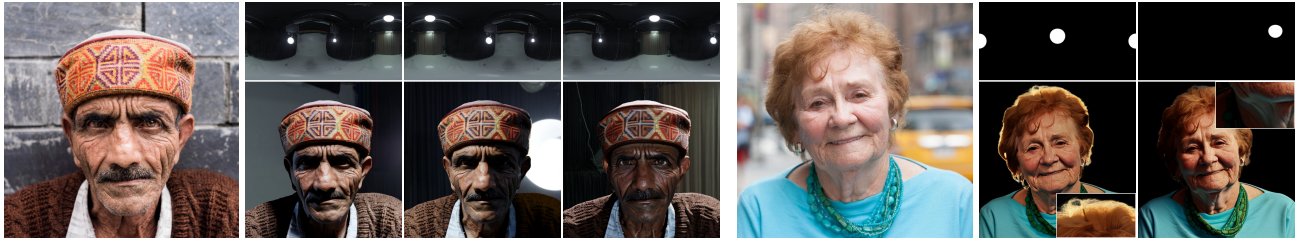
We start with a *Base* configuration that excludes both techniques. We then assess each method individually: *Base + Multi-Task* adds multi-task training, and *Base + Inference Adaptation* adds inference adaptation. Our full model,



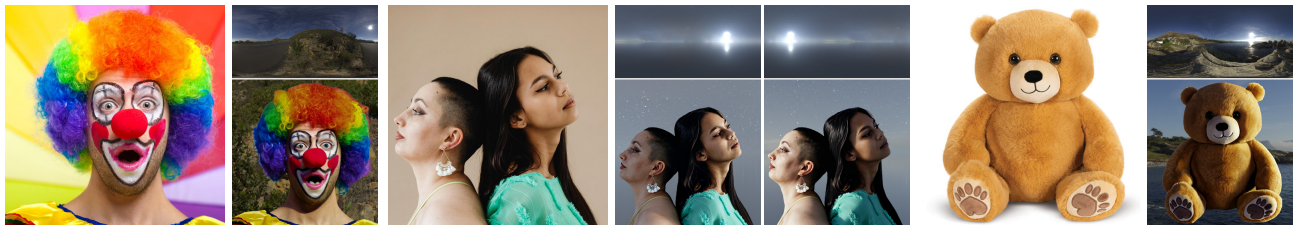
(a) Our method effectively relights subjects under outdoor (left) and indoor (right) illumination. Outdoors, it accurately renders cast shadows from facial features and glasses (see inset). Indoors, it captures complex lighting effects such as neon illumination.



(b) Our method captures interesting lighting effects for portraits, synthesizing fine details like catch light in the eye for realistic relighting (left, see inset) and subsurface scattering in the ear under strong backlight conditions, such as sunlight (right, see inset).



(c) Our method enables studio-style lighting for portraits, creating dramatic effects in studio settings (left). Using hand-crafted environment maps, we relight with two presets (right): Backlight, with a light behind the subject to define edges and create a rim effect (see inset); and Rembrandt, with angled lighting illuminating one side of the face and casting the other in shadow for depth and contrast, also revealing inter-reflections from clothing (rightmost, see inset).



(d) Our method generalizes to out-of-distribution images for relighting, such as image of a clown (portrait with heavy makeup), portrait of multiple people, and a teddy-bear (non-human subject).

Figure 6. Real-world results show our method’s ability to handle diverse lighting scenarios. Each example includes the input portrait (left), the environment map used for relighting (top right), and the relit output (bottom right).

combining both, is referred to as *Ours*. Finally, we examine the effect of adding light stage data to each training batch, denoted as *Ours + Light Stage*. See the supplementary for details on the light stage data.

Fig. 9 illustrates the effect of each configuration. We show the input image (leftmost), the environment map (top) and a 3D model rendered in Blender for reference (bottom). *Base* loses important details and fails to produce textures in clothing and accessories. Multi-task training in *Base + Multi-Task* helps partially recover details. *Base + Inference Adaptation* enhances finer details by leveraging informa-

tion present in the input but lacks photo-realism. Adding light stage data during training (*Ours + Light Stage*) addresses identity and texture issues but cannot remove strong shadows in the input while relighting. We conjecture this occurs because strong shadows are rare in light stage captures. The model also produces oversaturated images under strong sunlight (see supplementary). Similar artifacts appear in other light stage-trained methods (e.g., Switch-Light). Finally, *Ours* relights portraits with plausible lighting and preserves important details from the input portrait. These findings are supported by our evaluation in Tab. 3.



Figure 7. In-the-wild portrait results. DiLightNet [58] produces severe artifacts. IC-Light [60] does not relight due to background lighting condition. Neural Gaffer [18] generates inaccurate shadows on faces. Total Relighting [28] and SwitchLight [21] produce soft shadows even under strong sunlight and alter skin tones. Our method achieves superior relighting while also better preserving subject identity.

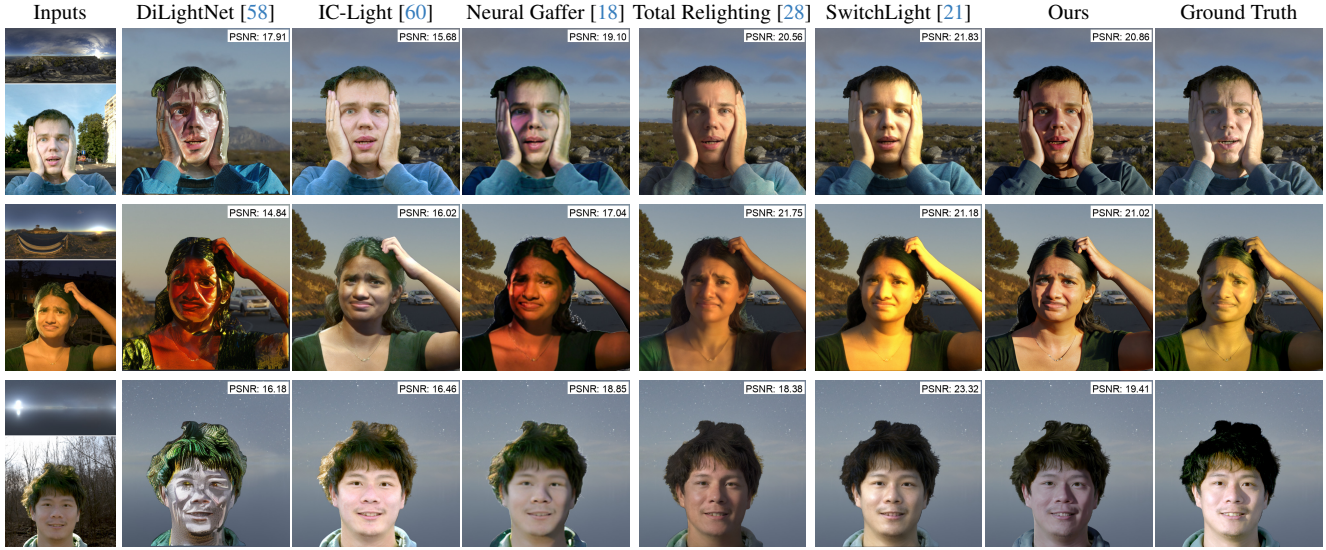


Figure 8. We compare of relighting the input portrait (bottom left) using environment map (top left) on a light stage test set.

4.5. Effect of Lighting Condition

We train two variants of our model: one conditioned on background and the other on the environment map for lighting. While the background-conditioned model often produces plausible lighting (e.g., strong cast shadows) and

appears well harmonized with the background, rotating the environment map reveals lighting inconsistencies (see Fig. 11). Although generating realistic shadows caused by self-occlusion is challenging for background-conditioned relighting methods such as IC-Light [60], leveraging our synthetic dataset enables us to address this effectively.

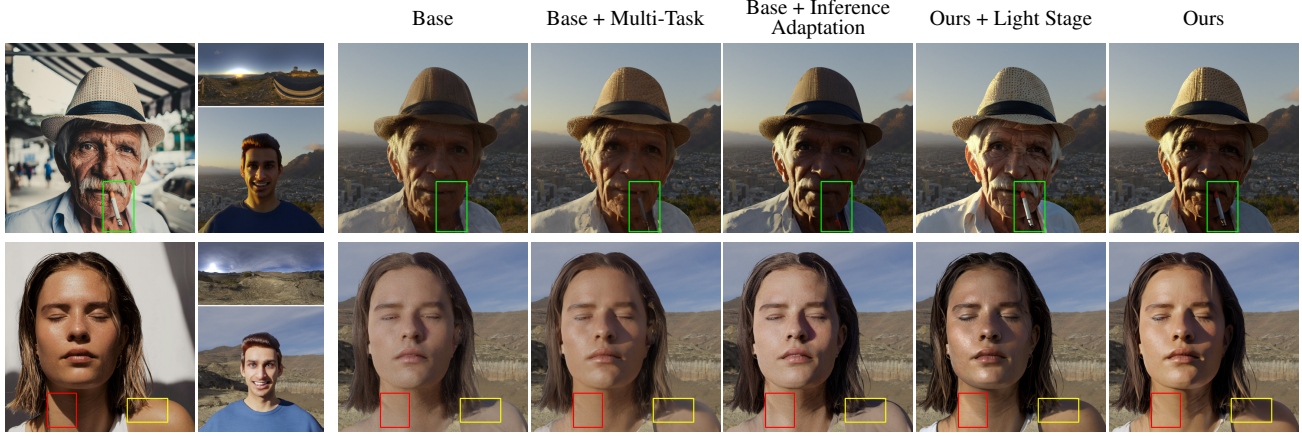


Figure 9. Training without our Multi-Task or Inference Adaptation techniques (*Base*) alters identity and loses textures. *Base + Multi-Task* recovers some details (e.g., skin tone, yellow box). *Base + Inference Adaptation* removes accessories (e.g., cigarette, green box) and generates unnatural skin textures (yellow box). *Ours* (*Base + Multi-Task + Inference Adaptation*) achieves plausible lighting effects while preserving portrait details. *Ours + Light Stage* improves details but fails to remove strong shadows from the input (red box).

	Base	Base+MT	Base+IA	Ours	Ours+LS	Ours+SD 1.5
Test Synthetic						
LPIPS _↓	0.066	0.066	0.062	<u>0.063</u>	0.065	0.069
SSIM _↑	0.942	0.942	0.946	<u>0.945</u>	0.942	0.937
PSNR _↑	29.13	29.05	29.64	<u>29.57</u>	29.13	28.30
FaceNet _↓	0.193	0.196	0.163	<u>0.165</u>	0.171	0.195
Test Light Stage						
LPIPS _↓	0.210	0.186	0.178	<u>0.165</u>	0.156	0.177
SSIM _↑	0.790	0.797	0.810	<u>0.813</u>	0.822	0.808
PSNR _↑	18.92	19.18	19.48	<u>19.70</u>	20.14	19.32
FaceNet _↓	0.295	0.242	0.179	<u>0.173</u>	0.149	0.188

Table 3. We evaluate the contribution of each component: Multi-Task training (*Base+MT*), inference adaptation (*Base+IA*), and their combination (*Ours*). Adding light stage data (*Ours+LS*) further improves performance on the light stage test set. Initializing with Stable Diffusion 1.5 (*Ours+SD 1.5*) yields slightly lower quantitative scores than *Ours* (initialized with IC-Light).

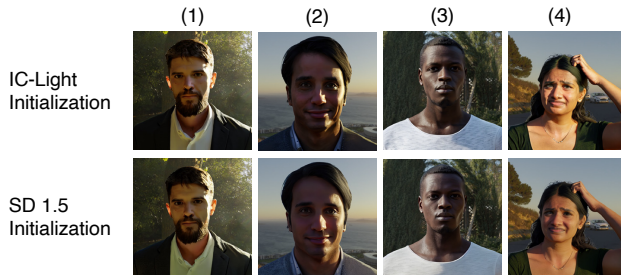


Figure 10. We compare SynthLight initialized with IC-Light [60] and with SD 1.5. Both yield similar visual quality and realistic lighting on in-the-wild subjects (col. 1–3) and the light stage set (col. 4), but IC-Light initialization performs better quantitatively.



Figure 11. Effect of Lighting condition. Limited lighting cues in background cause inaccurate shadow directions (e.g., (1)–(c)) compared to environment map. But synthetic data training still yields stronger, more plausible shadows than IC-Light [60].

5. Conclusion

We present SynthLight, a portrait relighting diffusion model capable of relighting in-the-wild portraits while garnering lighting supervision only from synthetic data. Our method demonstrates strong generalization capabilities, underscoring the potential of synthetic data to achieve plausible lighting effects such as strong cast shadows, realistic catch lights in the eyes, and natural inter-reflections. We discuss limitations and future work in the supplementary.

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